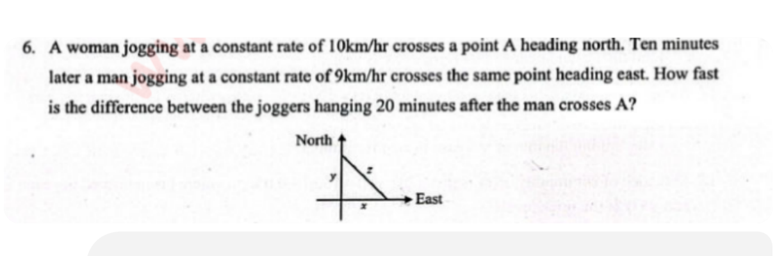


6. A woman jogging at a constant rate of 10km/hr crosses a point A heading north. Ten minutes later a man jogging at a constant rate of 9km/hr crosses the same point heading east. How fast is the difference between the joggers hanging 20 minutes after the man crosses A?

North
y
z
x
East

explain this quuestion in easy and short way

Is question ka matlab: **2 joggers A point se 90° angle (north & east) me bhag rahe hain**, aur humein dekhna hai **unke darmiyan distance kitni speed se badh raha hai** (rate of separation), us waqt jab **aadmi A cross karne ke 20 min baad**.

### Step 1: Time set karo

- Aadmi ne A **t = 0** par cross kiya.
- Aurat ne A **10 min pehle** cross kiya tha.

So jab aadmi ke **20 min** ho chuke:

- Aadmi ka time = 20 min = $\frac{1}{3}$ hr
- Aurat ka total time = 10 min + 20 min = 30 min = $\frac{1}{2}$ hr

### Step 2: Dono ne kitna distance cover kiya?

**Aadmi (east, 9 km/hr):**

$$x = 9 \times \frac{1}{3} = 3 \text{ km}$$

**Aurat (north, 10 km/hr):**

$$y = 10 \times \frac{1}{2} = 5 \text{ km}$$

### Step 3: Beech ka distance (Pythagoras)

$$z = \sqrt{x^2 + y^2} = \sqrt{3^2 + 5^2} = \sqrt{34}$$

### Step 4: Distance badhne ki speed (related rates)

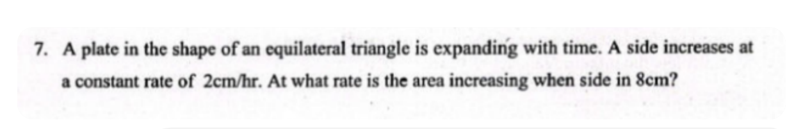
$$\frac{dz}{dt} = \frac{x\frac{dx}{dt} + y\frac{dy}{dt}}{z}$$

Yahan:

- $\frac{dx}{dt} = 9$
- $\frac{dy}{dt} = 10$
- $x = 3,\ y = 5,\ z = \sqrt{34}$

$$\frac{dz}{dt} = \frac{3 \cdot 9 + 5 \cdot 10}{\sqrt{34}} = \frac{27 + 50}{\sqrt{34}} = \frac{77}{\sqrt{34}} \approx 13.2 \text{ km/hr}$$

✅ **Answer: distance between them is increasing at about $13.2 \text{ km/hr}$.**



7. A plate in the shape of an equilateral triangle is expanding with time. A side increases at a constant rate of 2cm/hr. At what rate is the area increasing when side in 8cm?

explain it in easy and short way in english

We have an **equilateral triangle** whose side $s$ is growing.

- Given: $\frac{ds}{dt} = 2$ cm/hr
- Find: $\frac{dA}{dt}$ when $s = 8$ cm

### Step 1: Area formula

For an equilateral triangle:

$$A = \frac{\sqrt{3}}{4}s^2$$

### Step 2: Differentiate with respect to time

$$\frac{dA}{dt} = \frac{\sqrt{3}}{4} \cdot 2s \cdot \frac{ds}{dt} = \frac{\sqrt{3}}{2}s\frac{ds}{dt}$$

### Step 3: Plug in values $s = 8$, $\frac{ds}{dt} = 2$

$$\frac{dA}{dt} = \frac{\sqrt{3}}{2} \times 8 \times 2 = 16\sqrt{3}\ \text{cm}^2/\text{hr}$$

✅ **Answer:** $\boxed{16\sqrt{3}\ \text{cm}^2/\text{hr}}$

but area od triangle is 1/2 l*h

You're right: **area =** $\frac{1}{2}\,l\,h$ is always true.

But in an **equilateral triangle**, the height $h$ is not independent; it depends on the side $s$.

### For equilateral triangle:

Drop a height: it makes a 30-60-90 triangle, so

$$h = \frac{\sqrt{3}}{2}s$$

Now use your formula:

$$A = \frac{1}{2} \cdot s \cdot h = \frac{1}{2} \cdot s \cdot \frac{\sqrt{3}}{2}s = \frac{\sqrt{3}}{4}s^2$$

So it becomes the same formula we used.

### Now rate of change:

$$A = \frac{\sqrt{3}}{4}s^2 \Rightarrow \frac{dA}{dt} = \frac{\sqrt{3}}{2}s\frac{ds}{dt}$$

At $s = 8$, $\frac{ds}{dt} = 2$:

$$\frac{dA}{dt} = \frac{\sqrt{3}}{2} \cdot 8 \cdot 2 = 16\sqrt{3}\ \text{cm}^2/\text{hr}$$

✅ Final: $16\sqrt{3}\ \text{cm}^2/\text{hr}$

means for any triangle which formula we should use for area finding

For **any triangle**, you can use different area formulas depending on what information you have:

### 1) If you know base and height

$$A = \frac{1}{2}bh$$

✅ Most common, works for every triangle.

### 2) If you know all three sides $a, b, c$ (no height needed)

**Heron's formula:**

$$s = \frac{a+b+c}{2}, \quad A = \sqrt{s(s-a)(s-b)(s-c)}$$

### 3) If you know two sides and the included angle $C$

$$A = \frac{1}{2}ab\sin C$$

### 4) If the triangle is equilateral (special case)

$$A = \frac{\sqrt{3}}{4}s^2$$

**Rule of thumb:**

- Height given → use $\frac{1}{2}bh$
- 3 sides given → use **Heron**
- 2 sides + included angle → use $\frac{1}{2}ab\sin C$

one sie and 2 angle if

If you know **1 side + 2 angles**, you can still find the **area** easily.

### Steps (works for any triangle)

1. **Find the 3rd angle**

$$A = 180^\circ - (B + C)$$

2. **Use this direct area formula** (when side $a$ is the known side, and the other two angles are $B$ and $C$):

$$\boxed{\text{Area} = \frac{a^2\ \sin B\ \sin C}{2\ \sin A}}$$

(Angles in degrees.)

✅ This comes from: Sine rule + $\frac{1}{2}bc\sin A$

If your known side is $b$ or $c$ instead, same pattern:

$$\text{Area} = \frac{b^2 \sin A \sin C}{2 \sin B} \quad \text{or} \quad \text{Area} = \frac{c^2 \sin A \sin B}{2 \sin C}$$

If you want, tell me the **side value + two angles**, I'll compute the area.

Ship s2 is 20km due north of ship S2. what is mean by due north

**"Due north"** means **exactly north in a straight line** — **no east or west at all**.

So if Ship $S_2$ is **20 km due north** of Ship $S_1$, it means:

- $S_2$ is **directly above** $S_1$ on a map,
- distance between them is **20 km**,
- direction is **exactly 0° east/west** (just straight north).

if we say  north of west or west of north then

Those phrases mean the direction is **between** north and west (in the NW quarter). The words tell you **which main direction you start from**, and the other word tells you **which way you turn**.

✅ **"north of west"**

- Start facing **West**
- Turn **toward North**
- So it's a direction **between West and North**, closer to **West**
- Example: "30° north of west" = go west, then tilt 30° up toward north.

✅ **"west of north"**

- Start facing **North**
- Turn **toward West**
- So it's between North and West, closer to **North**
- Example: "30° west of north" = go north, then tilt 30° left toward west.

### Quick memory

- **"X of Y" = start at Y, turn toward X**
  So:
- **north of west**: start West → turn to North
- **west of north**: start North → turn to West

If you tell me the angle (like 25°), I can show it as a bearing too.

critical point in easy words

A **critical point** (easy words) is a point on a graph where the function **can stop going up or stop going down** — or where the slope is not normal.

### How to find it (simple rule)

A number $x = c$ is a **critical point** if:

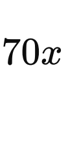
1. $f'(c) = 0$ (slope is **zero**, flat top/bottom), **or**
2. $f'(c)$ **does not exist** (corner/cusp/vertical tangent), **but** the function value exists.

### Why it matters

Critical points are where you can get:

- **maximum** (highest point),
- **minimum** (lowest point),
- or a **turning point**.

**Example:** $f(x) = x^2$
$f'(x) = 2x$. Set $2x = 0 \Rightarrow x = 0$.
So $x = 0$ is a critical point.



15. A company determines that for the production of $x$ units of a commodity its remove and cost functions are, respectively, $R(x) = -3x^2 + 970x$ and $G(x) = 2x^2 + 500$. Find the maximum profit and minimum average cost.

Solve it in easy way

Let

- Revenue: $R(x) = -3x^2 + 970x$
- Cost: $G(x) = 2x^2 + 500$

## 1) Maximum profit

Profit $P(x) = R(x) - G(x)$

$$P(x) = (-3x^2 + 970x) - (2x^2 + 500) = -5x^2 + 970x - 500$$

This is a **downward parabola** (because $-5 < 0$), so it has a **maximum** at its vertex:

$$x = \frac{-b}{2a} = \frac{-970}{2(-5)} = \frac{970}{10} = 97$$

Now profit at $x = 97$:

$$P(97) = -5(97^2) + 970(97) - 500$$

$97^2 = 9409$

$$P(97) = -5(9409) + 94090 - 500 = -47045 + 94090 - 500 = 46545$$

✅ **Maximum profit =** $46,545$ (when $x = 97$ units)

---

## 2) Minimum average cost

Average cost:

$$AC(x) = \frac{G(x)}{x} = \frac{2x^2 + 500}{x} = 2x + \frac{500}{x}$$

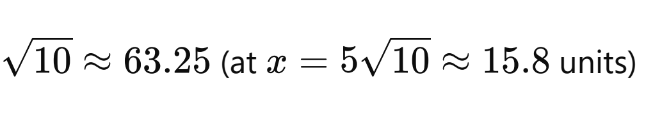
Differentiate and set to 0:

$$AC'(x) = 2 - \frac{500}{x^2} = 0 \Rightarrow 2 = \frac{500}{x^2} \Rightarrow x^2 = 250 \Rightarrow x = \sqrt{250} = 5\sqrt{10}$$

Now minimum average cost value:

$$AC_{\min} = 2(5\sqrt{10}) + \frac{500}{5\sqrt{10}} = 10\sqrt{10} + 10\sqrt{10} = 20\sqrt{10} \approx 63.25$$

✅ **Minimum average cost =** $20\sqrt{10} \approx 63.25$ (at $x = 5\sqrt{10} \approx 15.8$ units)



**Example 46:** Determine the interval on which the graph of $f(x) = -x^3 + \frac{9}{2}x^2$ is concave upward and the intervals for which the graph is concave downward.

Solve it

Given

$$f(x) = -x^3 + \frac{9}{2}x^2$$

### Step 1: Find the 2nd derivative

First derivative:

$$f'(x) = -3x^2 + 9x$$

Second derivative:

$$f''(x) = -6x + 9$$

### Step 2: Set $f''(x) = 0$ to find the change point

$$-6x + 9 = 0 \Rightarrow 6x = 9 \Rightarrow x = \frac{3}{2}$$

### Step 3: Test sign of $f''(x)$

- If $x < \frac{3}{2}$, say $x = 0$:

$$f''(0) = 9 > 0 \Rightarrow \text{concave up}$$

- If $x > \frac{3}{2}$, say $x = 2$:

$$f''(2) = -12 + 9 = -3 < 0 \Rightarrow \text{concave down}$$

✅ **Concave upward:** $(-\infty,\ \frac{3}{2})$
✅ **Concave downward:** $(\frac{3}{2},\ \infty)$

(Inflection point at $x = \frac{3}{2}$.)

relative extrema

**Relative extrema** means **local highest or local lowest points** of a graph (not necessarily the biggest/smallest overall).

### Types

- **Relative maximum (local max):** point where the function is higher than nearby points.
- **Relative minimum (local min):** point where the function is lower than nearby points.

### How to find relative extrema (easy steps)

1. Find derivative: $f'(x)$
2. Solve $f'(x) = 0$ (or where $f'(x)$ doesn't exist). These are **critical points**.
3. Test which type:
   - **Second derivative test:**
     - If $f''(c) > 0$ → **relative minimum**
     - If $f''(c) < 0$ → **relative maximum**
     - If $f''(c) = 0$ → test again (sign chart needed)

### Quick example

$f(x) = x^2$
$f'(x) = 2x = 0 \Rightarrow x = 0$
$f''(x) = 2 > 0$ so $x = 0$ is a **relative minimum**.